

ISC-matched average subjects for the multilingual *Le Petit Prince* fMRI dataset

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Abstract

Using individual fMRI scans from the *Le Petit Prince* dataset [2], we computed average subjects for English, French, and Chinese, carefully matching them in terms of number of participants ($n = 28$) and inter-subject correlations (ISC). The repository provides a) a brain mask common to all three languages; b) an average subject for each language; c) the inter-subject correlation maps for each language. The code and data are available at https://github.com/XinyuZhou2000/lpp_isc_matched_average_subjects. This allows one to perform comparative analyses on the *Le Petit Prince* dataset, controlling for data quality across languages.

Note that if you do not need to compare between the different languages, you might want to use the language-specific average subjects computed over the full pool of participants, available at https://github.com/l-bg/llms_brain_lateralization.

1 Background and motivation

In model–brain alignment studies which report correlations between a model’s predictions and fMRI time-series as brain score, one can rely either on individual data or on data averaged across multiple participants. In this work, we use fMRI data from the multilingual *Le Petit Prince* project. Here, for a given language, participants have been exposed to exactly the same stimuli, which allows to use average subjects, which can be appealing as it provides a better signal-to-noise ratio and limits the computations to one subject instead of many.

One issue with the *Le Petit Prince* fMRI dataset [2], is that participant pools are unbalanced across languages: 49 English participants, 35 Chinese participants, and 28 French participants. When average subjects are built from all available participants, the English average subject is based on substantially more participants than the French average subject, and has higher reliability. This difference could confound later comparisons of encoding performance across languages. Even if the number of subjects is matched, the quality of the measurements and the participants’ engagement in the task can lead to different ISC (see later Section 4). To reduce this confound, the analysis uses an ISC-matching procedure: all 28 French participants are retained, while 28 English and 28 Chinese participants are selected so that the resulting average subjects have comparable whole-brain ISC. As a diagnostic analysis, the study also examines how group size and comprehension quiz score affect ISC.

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ISC does not require a language model. It measures how similarly different participants’ brains respond to the same time-locked stimulus. In naturalistic story listening, participants hear the same narrative over time. If a voxel is reliable across subjects, its time series should be similar across independent groups of listeners.

2 Materials and Methods

2.1 Dataset and preprocessing

The multilingual *Le Petit Prince* fMRI dataset contains participants listening to the audiobook of *Le Petit Prince* in their native language [2]. The dataset includes English, French, and Chinese groups. Each participant was scanned during 9 runs, with a sampling period of 2 s.

The participant pools are unequal in size: 49 English participants, 35 Chinese participants, and 28 French participants. Since French is the smallest group, 28 participants were used as the target group size. In the final matched dataset, all 28 French participants were kept, while 28 English and 28 Chinese participants were selected from the larger pools.

First, using `nilearn.masking.compute_multi_epi_mask` with a threshold set to 0.5, we computed a common brain mask from all 112 participants. This mask contains 26,439 voxels and is distributed as `mask_lpp_common_en_cn_fr.nii.gz`.

Each subject’s fMRI timeseries were then extracted from this mask, using `nilearn’s NiftiMasker`, applying applied linear detrending, standardization and high-pass filtering with a cut-off of 1/128 Hz, and 8 mm full-width at half-maximum smoothing. Before computing correlations, we trimmed the time-series, deleting the first and last 10 samples (that is, 20s), as run boundaries sometimes contain transition artifacts. After this preprocessing, each language was represented by 9 run-level matrices with dimensions

$$N_{\text{subjects}} \times T_{\text{run}} \times V,$$

where N_{subjects} is the number of participants, T_{run} is the number of time points in the run after trimming, and V is the number of voxels in the common mask.

The preprocessing used for ISC matching and the preprocessing used for the final average subjects were closely related but not identical. Smoothing and trimming were used during ISC matching to obtain stable reliability estimates for group selection. After the matched groups were selected, the final average subjects were rebuilt with the preprocessing expected by the downstream encoding pipeline.

2.2 Group-wise inter-subject correlation

A split-half group-wise ISC procedure was used. For a given language and a given group of participants, the group was randomly divided into two equal halves. The fMRI signals were averaged within each half. Then, for each voxel, the Pearson correlation was computed between the two half-group average time series. This produced one voxel-wise ISC value. The whole-brain ISC score was obtained by averaging these voxel-wise values across voxels and across the 9 runs.

For one split s of a group G , the scalar ISC score was defined as

$$\text{ISC}_s(G) = \frac{1}{KV} \sum_{k=1}^K \sum_{v=1}^V r_{s,k,v}, \quad (1)$$

where $K = 9$ is the number of runs, V is the number of voxels, and $r_{s,k,v}$ is the correlation between the two half-group average time series for run k and voxel v .

A high ISC means that two independent halves of the group show similar time courses. A low ISC means that the group-level response is noisier or less consistent across participants.

Note that this analysis follows the general average-subject framework of Bonnassee-Gahot and Pallier [1], but uses a simpler reliability estimate, which directly computed via a voxel-wise correlation between the two groups.

2.3 Selecting participants to create ISC-matched average subjects

The downstream encoding analyses fit language-model predictors to fMRI time series. For each language and each run, the fMRI time series are averaged across a selected group of participants. This produces one group-level fMRI target per language. The average subjects must be constructed in a comparable way. If one language average is based on more participants, or on a group with higher inter-subject reliability, it may be easier to predict.

For this reason, the English, Chinese, and French average subjects were not built from all available participants. Instead, ISC-matched average subjects were constructed. The goal was to use the same number of participants in each language and to match the inter-subject correlation of the group-level fMRI response before running encoding analyses.

Group selection

French had the smallest participant pool, with 28 participants, and was therefore used as the reference group. All 28 French participants were kept. English and Chinese had larger participant pools, with 49 and 35 participants respectively. For these two languages, 28 participants were selected from the larger pools.

The selection was based on repeated split-half ISC. For English and Chinese, 50 candidate groups of 28 participants were randomly sampled. Each candidate group was evaluated with 10 repeated random split-half estimates. In each split, the 28 participants were divided into two halves of 14 participants. The fMRI signals were averaged within each half, and voxel-wise ISC was computed between the two half-group average time series. The voxel-wise ISC values were then averaged across voxels and runs to obtain one scalar ISC score for that split. Finally, the 10 split scores were averaged to obtain a stable ISC estimate for the candidate group.

The fixed French group was evaluated using the same repeated split-half ISC procedure. This made the French reference score comparable to the English and Chinese candidate scores. This repeated procedure was used because the ISC of a group should not depend on one arbitrary split. A single split can give a noisy estimate of group reliability. Repeating the split several times gives a more stable score and makes the group selection more robust.

Matching criterion

The selected English and Chinese groups were chosen to match the fixed French group in whole-brain ISC. For each candidate group, the stable ISC score was defined as the average of its repeated split-half estimates:

$$\text{ISC}(G) = \frac{1}{S} \sum_{s=1}^S \text{ISC}_s(G), \quad (2)$$

where $S = 10$ is the number of repeated splits.

The final English–Chinese pair was selected by minimizing the range of the three language scores:

$$\Delta = \max(\text{ISC}_{\text{EN}}, \text{ISC}_{\text{CN}}, \text{ISC}_{\text{FR}}) - \min(\text{ISC}_{\text{EN}}, \text{ISC}_{\text{CN}}, \text{ISC}_{\text{FR}}). \quad (3)$$

Table 1: Summary statistics of voxel-wise ISC values for the final matched groups. The table reports the mean, median, and maximum voxel-wise ISC values for the selected English, Chinese, and French groups.

Language	Mean	Median	Max
English	0.2983	0.2711	0.7778
Chinese	0.2988	0.2663	0.8440
French	0.2991	0.2752	0.8088

This criterion directly follows the purpose of the procedure: the three average subjects should have similar whole-brain reliability.

2.4 Construction of an average subjects

After selecting the matched groups, the average subjects were rebuilt from the selected participant lists. For each language and each run, the selected participants’ fMRI time series were transformed with the common mask. The average-subject script applied detrending, standardization, and high-pass filtering with a 1/128 Hz cut-off.

For each language l and each run k , the average subject was computed as the voxel-wise mean of the 28 selected participants:

$$\bar{Y}_{l,k} = \frac{1}{|S_l|} \sum_{i \in S_l} Y_{i,l,k}, \quad (4)$$

where S_l is the selected participant set for language l , and $Y_{i,l,k}$ is the fMRI time series of participant i in language l and run k .

The resulting average time series was standardized again across time for each voxel and saved as a compressed matrix. The output consisted of one average subject per language, with 9 run-level files for each language. The selected participant IDs used to construct these matched average subjects are listed in Appendix B. This final reconstruction step ensured that the average subjects used by the downstream encoding pipeline were generated with the expected preprocessing choices, while the participant lists themselves were selected using reliability matching.

3 Results of ISC matching

The final selected groups had highly similar reliability profiles. At the scalar level, the selected English, Chinese, and French groups were matched to have nearly identical whole-brain ISC scores. To further inspect whether this scalar matching was also reflected at the voxel level, the distribution of voxel-wise ISC values was examined in the selected groups.

The mean voxel-wise ISC values were 0.2983 for English, 0.2988 for Chinese, and 0.2991 for French (Table 1). The medians were also close across languages: 0.2711, 0.2663, and 0.2752, respectively. This indicates that the three groups were well matched in overall reliability.

A glass-brain visualization of the voxel-wise inter-subject reliability maps can also be used as a qualitative check. In such maps, color intensity represents voxel-wise ISC values, and dotted contours can mark the top 25% most reliable voxels within each language. The similarity between the three languages is not limited to a single scalar score. The voxel-wise ISC distributions also largely overlap across languages (Figure 1), suggesting that the matched groups provide comparable fMRI targets for the following encoding analyses. As an additional qualitative check, (Figure 2) shows glass-brain visualizations of the voxel-wise ISC maps.

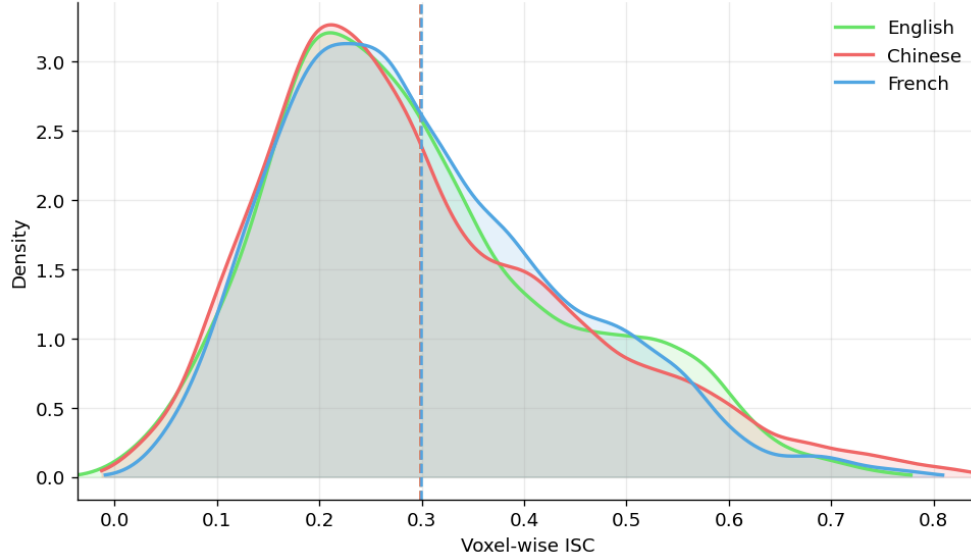


Figure 1: Voxel-wise ISC profiles of the final matched groups. The final English, Chinese, and French groups contain 28 participants each and have nearly identical scalar ISC values. The voxel-wise ISC distributions also largely overlap across languages.

4 Exploratory ISC analyses

Two factors that could affect group-wise ISC were examined. The first factor was sample size: larger groups should produce more reliable average responses because individual noise is reduced by averaging. The second factor was comprehension quiz score: participants who understood the story better may show more synchronized neural responses during naturalistic listening.

4.1 Effect of sample size

The first analysis tested whether ISC increases when more participants are averaged. For each language, random groups were sampled from the full participant pool. The half-group sizes were $n = 10$, $n = 12$, and $n = 14$, corresponding to total group sizes of 20, 24, and 28 participants. For each language and each group size, 30 independent resampling iterations were run with controlled random seeds. For each resample, participants were split into two halves, the two half-group responses were averaged, and voxel-wise ISC was computed across the 9 runs.

Whole-brain ISC increased with group size in all three languages (Figure 3). In Chinese, ISC increased from 0.246 at $n = 10$ to 0.304 at $n = 14$. In English, ISC increased from 0.223 at $n = 10$ to approximately 0.28 at $n = 14$. French showed the same trend and reached approximately 0.29 at the largest group size.

This result confirms that averaging more participants reduces individual noise and produces a more stable group-level response. It directly motivates the matched-average-subject procedure. If participant number affects ISC, then comparing average subjects built from different numbers of participants would introduce a confound. A language with more participants would have a more reliable target, and this could inflate its model-brain alignment score.

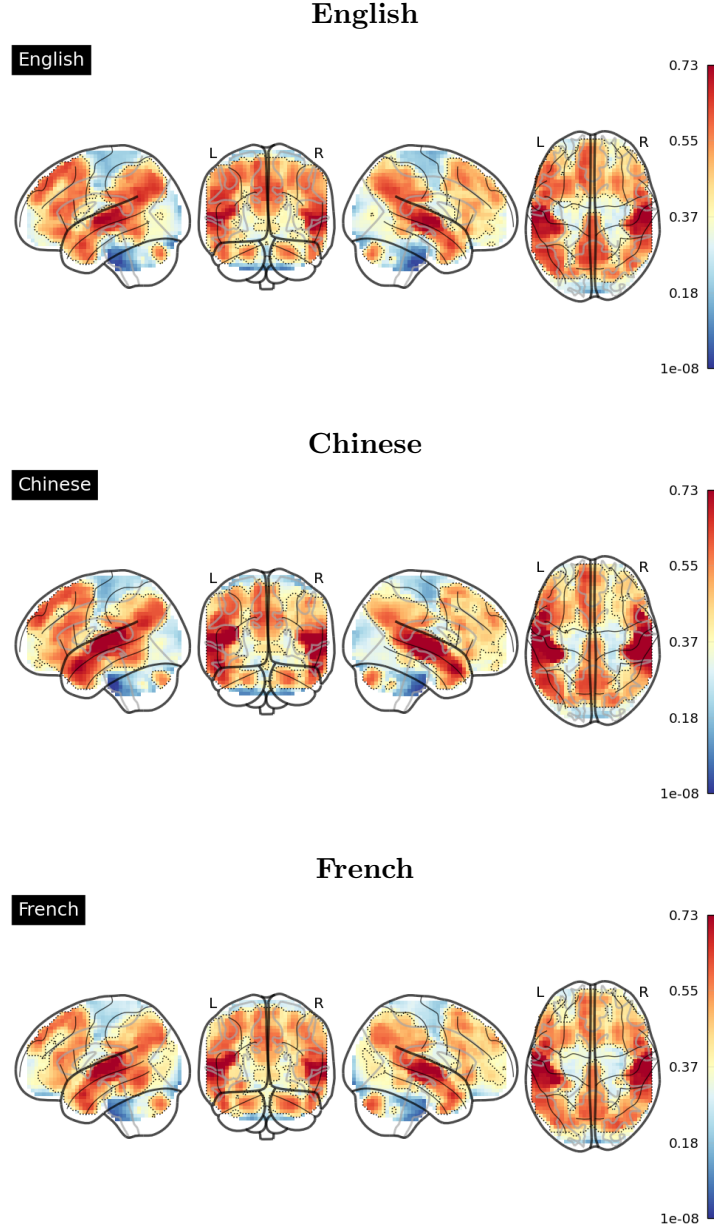


Figure 2: Glass-brain representation of voxel-wise inter-subject reliability. The three panels show voxel-wise ISC maps for the final selected English, Chinese, and French groups. Hotter colors indicate higher inter-subject reliability. Dotted contours mark the top 25% most reliable voxels within each language-specific ISC map.

4.2 Effect of comprehension quiz score

The second exploratory analysis tested whether participants with higher comprehension quiz scores also showed higher ISC. This analysis was conducted for English and Chinese, because comprehension quiz scores were available for these two languages. French was not included in this analysis because comparable comprehension quiz metadata were not available.

For each language, participants were ranked by the number of correctly answered comprehension questions. Groups were then constructed from the top-scoring participants, the

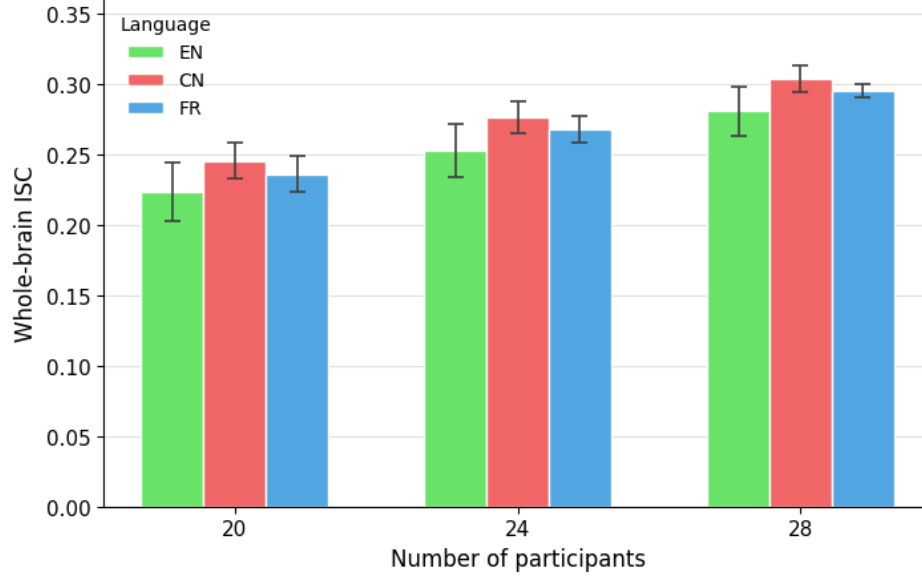


Figure 3: Effect of participant number on whole-brain ISC. For each language, random groups were sampled with total group sizes of 20, 24, and 28 participants. ISC was computed using split-half group-wise correlations. Whole-brain ISC increased with participant number in English, Chinese, and French.

bottom-scoring participants, and random reference groups. ISC was computed for each group using the same split-half procedure as above.

In English, the top-score groups showed higher ISC than the bottom-score groups across all group sizes (Figure 4). The difference was especially visible at smaller group sizes. As group size increased, the gap between top-score and bottom-score groups became smaller, but the top-score groups still showed the highest reliability.

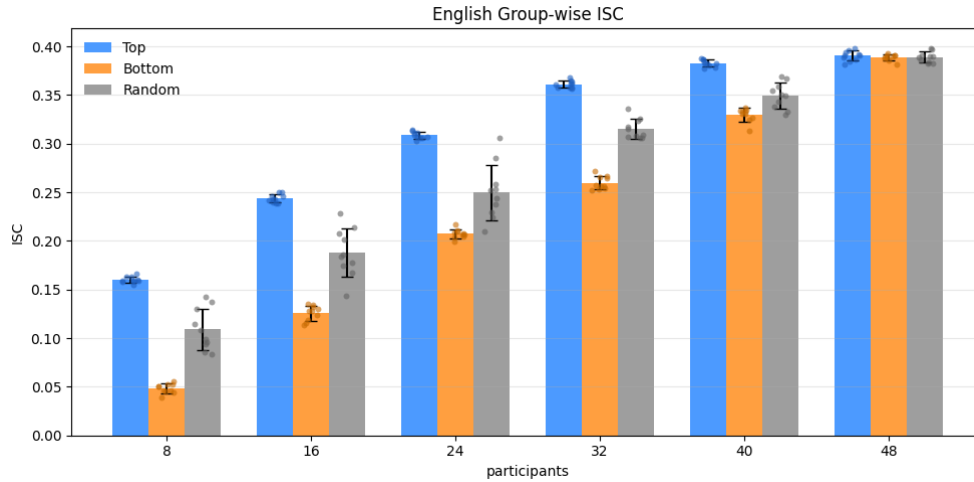


Figure 4: Effect of comprehension quiz score on English ISC. English participants were ranked by the number of correctly answered comprehension questions. For each group size, ISC was computed for the top-score group, the bottom-score group, and a random reference group.

A similar pattern was observed in Chinese (Figure 5). Top-score groups generally showed higher ISC than bottom-score groups, especially at smaller group sizes. Because the Chinese pool contained only 35 participants, this analysis could not be extended to the larger English group sizes of 40 and 48 participants.

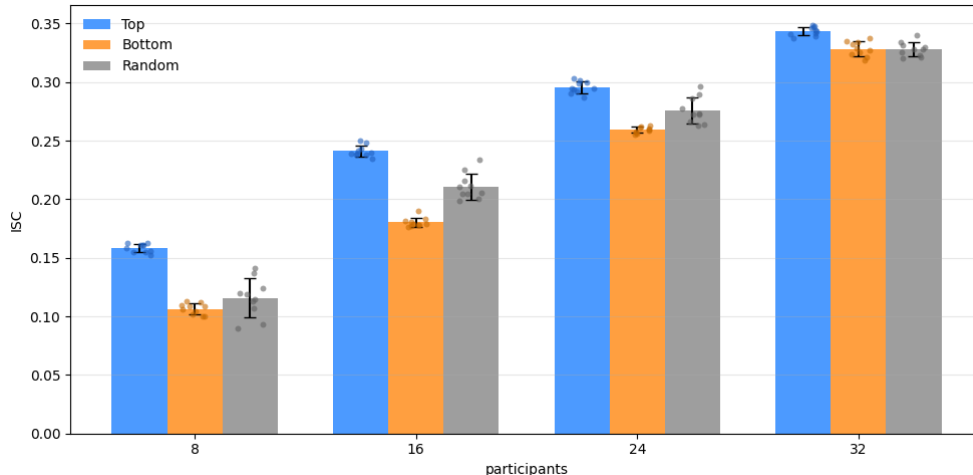


Figure 5: Effect of comprehension quiz score on Chinese ISC. Chinese participants were ranked by the number of correctly answered comprehension questions. For each group size, ISC was computed for the top-score group, the bottom-score group, and a random reference group.

These results suggest that ISC is related not only to the number of participants being averaged, but also to behavioral comprehension during the task. Participants who answered more comprehension questions correctly tended to show more synchronized neural responses, especially when the group size was small. This analysis was exploratory and was not used as the direct criterion for the final group matching; the final selection was based on whole-brain ISC. The full numerical summaries of the English and Chinese comprehension-score analyses are provided in Appendix A.

5 Summary

This report described the ISC and average-subject pipeline used to construct comparable fMRI targets for English, Chinese, and French in the multilingual *Le Petit Prince* dataset.

The exploratory analyses showed that group-wise ISC increases with sample size and is affected by listener comprehension. These results confirmed that the reliability of an average brain depends on the participants included in the group. Therefore, directly comparing average subjects built from unequal participant pools would introduce a methodological confound.

The final ISC-matching procedure controlled this issue by constructing matched average subjects for English, Chinese, and French. French was fixed at 28 participants, while English and Chinese were selected from larger pools to match the French whole-brain ISC. The final selected groups had very similar scalar ISC values and largely overlapping voxel-wise ISC distributions.

The main conclusion is methodological: before comparing model–brain alignment across languages, the brain targets must be made comparable. ISC-matched average subjects provide a practical way to control participant number and whole-brain fMRI reliability, making subsequent cross-language encoding analyses more interpretable.

A Comprehension quiz analysis

The comprehension-score analyses were used as exploratory checks of whether behavioral comprehension was associated with group-wise ISC. Participants were ranked by the number of correctly answered comprehension questions. For each group size, ISC was computed for top-score groups, bottom-score groups, and random reference groups. French was not included because comparable comprehension quiz metadata were not available.

Table A.1: English comprehension quiz analysis. Participants were ranked by the number of correctly answered comprehension questions. Values show mean whole-brain ISC $\pm$ SD across resampling iterations.

Half size n	Total size $2n$	Top ISC	Bottom ISC	Random ISC
4	8	0.160 ± 0.003	0.048 ± 0.005	0.109 ± 0.021
8	16	0.244 ± 0.004	0.126 ± 0.008	0.189 ± 0.025
12	24	0.309 ± 0.004	0.208 ± 0.005	0.250 ± 0.029
16	32	0.361 ± 0.004	0.260 ± 0.007	0.315 ± 0.010
20	40	0.383 ± 0.004	0.330 ± 0.007	0.350 ± 0.014
24	48	0.391 ± 0.005	0.389 ± 0.003	0.389 ± 0.006

Table A.2: Chinese comprehension quiz analysis. Participants were ranked by the number of correctly answered comprehension questions. Values show mean whole-brain ISC $\pm$ SD across resampling iterations.

Half size n	Total size $2n$	Top ISC	Bottom ISC	Random ISC
4	8	0.159 ± 0.004	0.106 ± 0.005	0.116 ± 0.017
8	16	0.241 ± 0.005	0.180 ± 0.004	0.211 ± 0.011
12	24	0.296 ± 0.005	0.259 ± 0.003	0.276 ± 0.011
16	32	0.344 ± 0.004	0.329 ± 0.006	0.328 ± 0.006

B Matched participant IDs

B.1 English

The selected English participants were:

sub-EN057	sub-EN058	sub-EN061	sub-EN063
sub-EN064	sub-EN065	sub-EN067	sub-EN070
sub-EN072	sub-EN075	sub-EN078	sub-EN079
sub-EN081	sub-EN083	sub-EN084	sub-EN086
sub-EN087	sub-EN091	sub-EN092	sub-EN096
sub-EN097	sub-EN098	sub-EN100	sub-EN103
sub-EN105	sub-EN109	sub-EN113	sub-EN114

B.2 Chinese

The selected Chinese participants were:

sub-CN001	sub-CN002	sub-CN003	sub-CN006
sub-CN007	sub-CN008	sub-CN009	sub-CN010
sub-CN011	sub-CN013	sub-CN014	sub-CN015
sub-CN017	sub-CN020	sub-CN021	sub-CN022
sub-CN023	sub-CN024	sub-CN025	sub-CN026
sub-CN028	sub-CN030	sub-CN031	sub-CN032
sub-CN033	sub-CN034	sub-CN036	sub-CN037

B.3 French

The selected French participants were:

sub-FR001	sub-FR002	sub-FR003	sub-FR004
sub-FR005	sub-FR006	sub-FR007	sub-FR008
sub-FR009	sub-FR010	sub-FR011	sub-FR012
sub-FR013	sub-FR014	sub-FR015	sub-FR016
sub-FR017	sub-FR018	sub-FR019	sub-FR020
sub-FR022	sub-FR023	sub-FR024	sub-FR025
sub-FR026	sub-FR028	sub-FR029	sub-FR030

References

- [1] Bonnasse-Gahot, L. and Pallier, C. (2024). fMRI predictors based on language models of increasing complexity recover brain left lateralization. *Advances in Neural Information Processing Systems*, 37.
- [2] Li, J., Bhattasali, S., Zhang, S., Franzluebbers, B., Luh, W.-M., Spreng, R. N., Brennan, J. R., Yang, Y., Pallier, C., and Hale, J. (2022). Le Petit Prince multilingual naturalistic fMRI corpus. *Scientific Data*, 9, Article 530. <https://doi.org/10.1038/s41597-022-01625-7>.